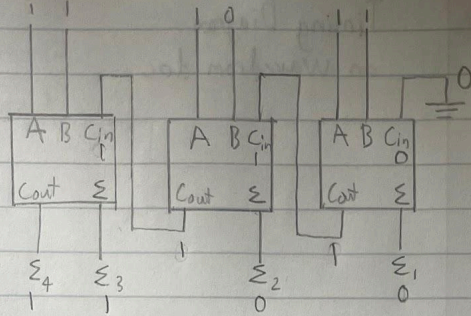


Loren
Grey

Problem Set #6

4.

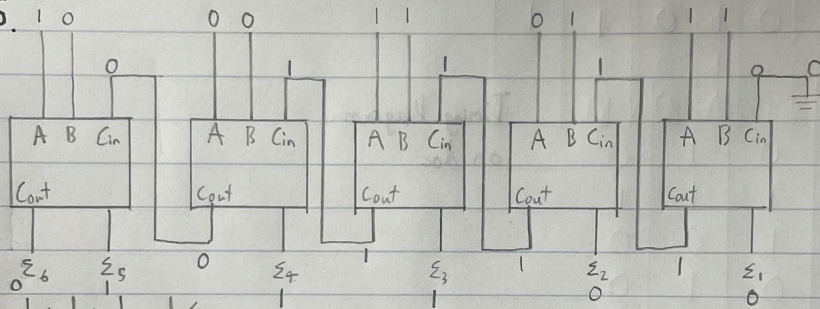


Cin	A	B	Cout	Σ
0	1	1	1	0
1	0	1	0	0
1	1	1	1	1

$$\begin{array}{r} 111 \\ + 0111 \\ + 0101 \\ \hline 1100 \end{array}$$

= 1100

5.

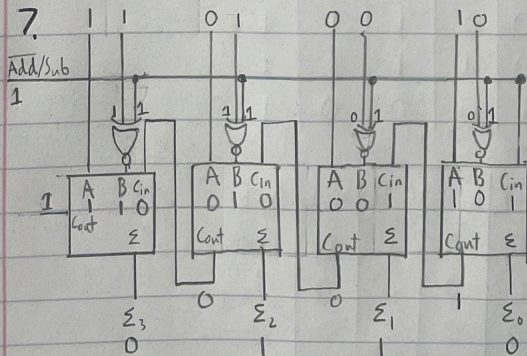


Cin	A	B	Cout	Σ
0	1	1	1	0
1	0	1	1	0
1	1	1	1	1
1	0	0	0	1
0	1	0	0	1

$$\begin{array}{r} 1111 \\ + 10101 \\ + 00111 \\ \hline 11100 \end{array}$$

= 011100

7.



Cin	A	B	Cout	Σ
1	1	0	1	0
1	0	0	0	1
0	0	1	0	1
0	1	1	1	0

= 0110

8.	C_{in}	A_1	B_1	Σ_1	A_2	B_2	Σ_2	C_{out1}	C_{out2}
	0	1	1	0	0	0	0	1	0
	0	1	1	0	1	1	0	1	1
	0	0	0	0	1	0	1	0	0
	0	0	0	0	0	1	1	0	0
	1	0	0	1	1	0	0	0	1
	1	1	0	0	1	1	1	1	1
	1	1	1	1	1	0	0	1	1
	1	1	1	1	0	1	0	1	1

Timing Diagram
on wavedrom doc

13.	A_0	B_0	A_1	B_1	$A_0=B_0$	$A_1=B_1$	$A=B$
	1	0	1	1	0	1	0
	0	0	0	0	1	1	1
	1	1	1	0	1	0	0
	0	1	0	0	0	1	0
	1	1	0	1	1	0	0

Timing Diagram
on doc

14.

A_0	A_1	A_2	A_3	B_0	B_1	B_2	B_3
1	0	0	1	0	0	1	0
1	1	1	1	1	1	1	1
0	1	1	1	0	1	0	0
0	0	1	1	1	1	0	0
1	0	1	0	0	0	1	1

$A_0 > B_0$	$A_1 > B_1$	$A_2 > B_2$	$A_3 > B_3$	$A > B$
1	0	0	1	1
0	0	0	0	0
0	0	1	1	1
0	1	1	1	1
1	0	0	0	1

Timing Diagram
on Doc

$A_0=B_0$	$A_1=B_1$	$A_2=B_2$	$A_3=B_3$	$A=B$
0	1	0	0	1
1	1	1	1	1
1	1	0	0	1
0	0	0	0	0
0	1	1	0	1

$A_0 < B_0$	$A_1 < B_1$	$A_2 < B_2$	$A_3 < B_3$	$A < B$
0	0	1	0	1
0	0	0	0	0
0	0	0	0	0
1	0	0	0	1
0	0	0	1	1

19. A | B | C | D

1 | 0 | 1 | 0

1 | 1 | 0 | 0

0 | 0 | 0 | 1

1 | 0 | 1 | 1

$$\bar{A}\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + A\bar{B}C\bar{D}$$

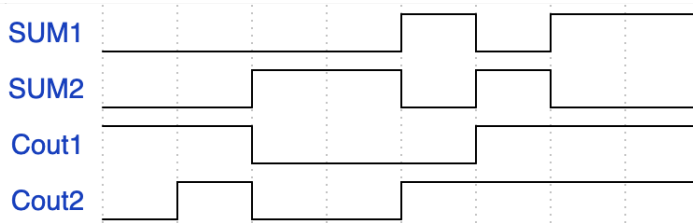
$$= \bar{A}\bar{B}C(\bar{D} + D) + A\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D$$

$$= \boxed{\bar{A}\bar{B}C + A\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D}$$

Logic circuit
on Doc

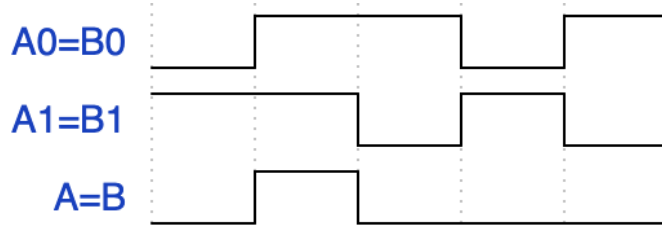
8.

```
{ signal: [  
  { name: "SUM1", wave: 'l...h|h.' },  
  { name: "SUM2", wave: 'l.h.l|l.' },  
  { name: "Cout1", wave: 'h.l..h..' },  
  { name: "Cout2", wave: 'l|l.h...' }  
]}
```

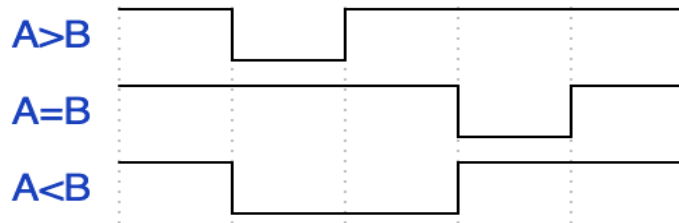


13.

```
{ signal: [  
  { name: "A0=B0", wave: 'l|h.l|h' },  
  { name: "A1=B1", wave: 'h.l|h|' },  
  { name: "A=B", wave: 'l|h|..' },  
]}
```



```
{ signal: [
  { name: "A>B", wave: 'hlh..' },
  { name: "A=B", wave: 'h..lh' },
  { name: "A<B", wave: 'hl.h.' },
]
```



```
{ assign:[
  ["out",
    ["|",
      ["&", "A",["&", ["~", "B"],"C"]],
      ["|", ["&", "A",
        ["&", "B",["&",
          ["~", "C"],["~", "D"]]]],
        ["&", ["~", "A"],["&", ["~", "B"],["&", ["~", "C"],"D"]]]]]
    ]
  ]
}]
```

